

Name: _____ Student ID: _____

1. (8 points) Expand the following logarithmic functions as much as possible:

(a) $\ln(5x^2y)$

(b) $\ln(\sqrt{xy}/z^3)$

(c) $\ln\left(6x\sqrt{y^2 + y/z}\right)$

(d) $\ln(e^{2x}y^3)$

2. (8 points) Take the first partial derivatives of the following functions **with respect to y**:

(a) $f(x, y) = \ln(x^5) + y^2/x$

(b) $f(x, y) = e^{x^{1/3}y^{1/3}}$

(c) $f(x, y) = 3x^2y - 2xy^3 + 4y$

(d) $f(x, y) = \ln(x^2 + y^2)$

3. (12 points) Find the critical points (if any) for the following function. For each critical point determine whether the function is at a local minimum or maximum.

$$f(x) = 5xe^{-x}$$

4. (12 points) Consider the function $f(x, y) = xy$. Find the values x^* and y^* that maximize f when x and y must sum to 1. (Hint: use Lagrange and think of 1 as income in a budget constraint.)

Bonus: Name an economist who won a Nobel Prize this week.